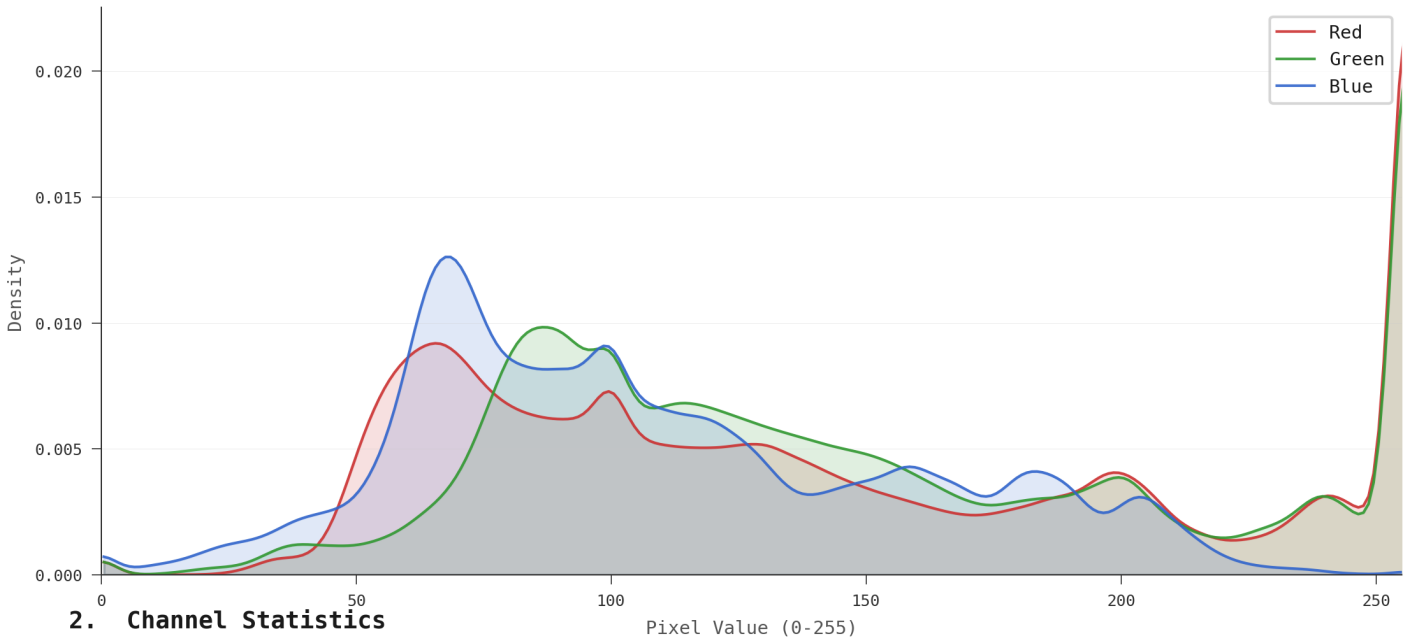
# Reference Data Sheet: KODIM06

Subject: Boat Zentime in Harbor

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

## 1. RGB Channel Distribution




## 2. Channel Statistics

| Channel | Mean | Std Dev | Variance | Kurtosis | Skewness | Min | Max |
|---|---|---|---|---|---|---|---|
| Red | 133.1676 | 65.6483 | 4309.6945 | -0.9540 | 0.5242 | 0 | 255 |
| Green | 139.3122 | 60.2086 | 3625.0755 | -0.7381 | 0.5106 | 0 | 255 |
| Blue | 108.7812 | 48.5409 | 2356.2226 | -0.5530 | 0.5034 | 0 | 255 |

## 3. Inter-Channel Correlation

| | R | G | B |
|---|---|---|---|
| R | 1.0000 | 0.9772 | 0.9603 |
| G | 0.9772 | 1.0000 | 0.9911 |
| B | 0.9603 | 0.9911 | 1.0000 |

**Pairwise Pearson Correlation Coefficients**

| | | |
|---|---|---|
| r(R-G) | = | **0.977168** |
| r(R-B) | = | **0.960329** |
| r(G-B) | = | **0.991059** |
| Avg \|r\| | = | **0.976185** |

## 4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{4309.7055} & 3862.3529 & 3060.2200 \\
3862.3529 & \mathbf{3625.0847} & 2896.4590 \\
3060.2200 & 2896.4590 & \mathbf{2356.2285}
\end{bmatrix}
\end{array}
$$

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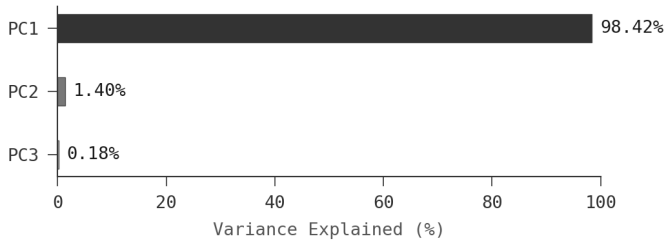
## 5. Eigendecomposition




**Eigenvector Loadings**

| | R | G | B | Eigenvalue |
|---|---|---|---|---|
| **PC1** | **-0.6462** | -0.5960 | -0.4766 | 10128.6724 |
| **PC2** | -0.7417 | 0.3436 | 0.5760 | 143.6789 |
| **PC3** | 0.1796 | -0.7257 | 0.6642 | 18.6675 |

## 6. Derived Metrics

**Condition Number**

**542.58**

(l1/l3) PC1/PC2: 70.50 | PC2/PC3: 7.70

**Blue Channel Independence**

**2.4%**

Fraction of blue variance not captured by PC1

**PC1 Dominant Channel**

**Red**

Loading: -0.6462 | Red dominant

**Classification: Near-Degenerate (PC1 = 98.42%)**

## 7. Spatial Autocorrelation (lag-1)

| Channel | Horizontal | Vertical | Average |
|---|---|---|---|
| Red | 0.979433 | 0.946378 | 0.962905 |
| Green | 0.974956 | 0.937302 | 0.956129 |
| Blue | 0.963681 | 0.911326 | 0.937503 |

## 8. Average Local Variance (3x3)

| Channel | Value |
|---|---|
| Red | 244.0803 |
| Green | 241.0629 |
| Blue | 213.1997 |

## 9. Redundancy Profile

**Classification:** Near-Degenerate | Condition Number: 542.58 | PC1: 98.42%

**Correlation:** G–B pair dominates at r = 0.991059 | R–G: 0.977168 | R–B: 0.960329

**Blue Channel:** 2.4% independent | 97.6% captured by PC1

**Spatial:** Avg autocorrelation: 0.952179 | Avg local variance: 232.7810

**Eigenvector:** Red dominant | R (-0.6462) leads | G (-0.5960), B (-0.4766)

## References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

*Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100*

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."